Because triangles Pxv and SPM are similar, and the parabola's focal property makes the sides SP and SM equal, the corresponding sides Px (that is QR) and Pv are also equal — $QR = Pv$. Now invoke the conic sections: the square of the ordinate equals the rectangle of the latus rectum and Pv. By Lemma XIII the parabola's latus rectum is $4PS$, so $Qv^2 = 4PS \cdot Pv = 4PS \cdot QR$. The curve's geometry has collapsed cleanly into the focal distance.